# Mathematics for Social Scientists

## Mid-course Review

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# Outline

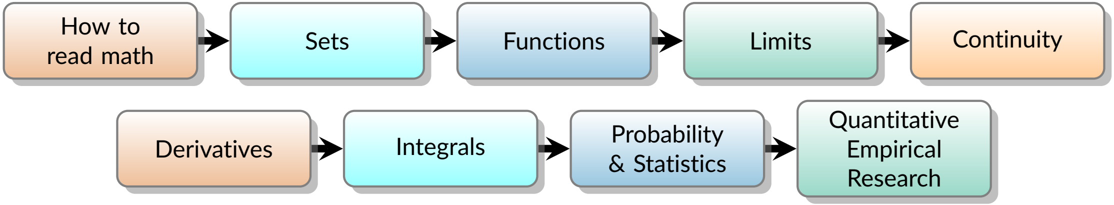
# Definitions

**Analysis.** The study of functions—how outputs behave as inputs vary and as values "approach" others. In plain terms: a careful way to say when "getting closer," "no sudden jumps," and "adding up tiny pieces" actually make sense. It tells us when a curve behaves smoothly and when totals we compute by slicing things finely are real and reliable.

**Calculus.** The study of continuous change. Practically, it answers two questions: "How fast is this changing right now?" and "How much has this added up to?" The first answer is the derivative (the best local slope); the second is the integral (adding up lots of tiny bits). They are two sides of the same coin.

**Linear Algebra.** The study of vectors and the linear rules that act on them. Think of many numbers handled as one object, combined or scaled in consistent ways. It provides directions, coordinates, and principal directions, so you can represent data cleanly, spot patterns, and describe how systems push information around.

# Analysis and Calculus: Building blocks



# Applications in the Social Sciences I

**Sets**

- Data wrangling as union/intersection/difference
- Defining populations, samples, frames, treatment groups
- Record linkage and de-duplication as set matching

**Functions**

- Modeling relations $y = f(x)$ (policy $\rightarrow$ outcome)
- Pipelines via composition (exposure $\rightarrow$ behavior $\rightarrow$ outcome)
- Utility/production curves; response surfaces

**Continuity**

- Small input changes yield small output changes (stability)
- Assumptions behind RDD (regression discontinuity design) and smooth counterfactuals

**Derivatives**

- Rates, elasticities, marginal effects
- Optimization for choices and policy design (FOC/SOC)

# Applications in the Social Sciences II

**Integrals**

- Aggregating flows to totals (e.g., turnout over time)
- Area-under-curve; probability mass from densities
- Expected values as weighted totals

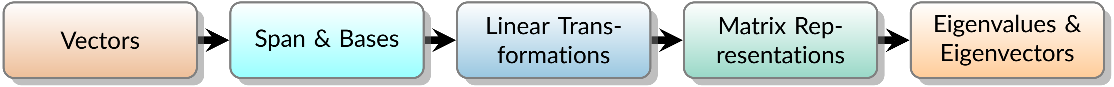
**Probability & Statistics**

- Quantifying uncertainty; Bayes' rule; likelihoods
- Sampling distributions, standard errors, confidence sets

**Quantitative Empirical Research**

- Estimation (regression, GLMs) and diagnostics
- Design-based and model-based causal analysis
- Forecasting and policy simulation

# Linear Algebra: Building Blocks



# Applications in the Social Sciences III: Linear Algebra

**Representation & estimation**

- Data tables/design matrices; OLS as projection onto the column space of $X$
- Fixed effects: within-transformation (demeaning) as a projection
- Instrumental variables / 2SLS as sequential projections

**Dimension reduction**

- PCA/SVD to build composite indices (e.g., socioeconomic status)
- Factor models and latent-trait measurement (e.g., IRT)

**Networks & structure**

- Eigenvector centrality/PageRank from adjacency matrices
- Diffusion and contagion via powers of $A$; Laplacian methods

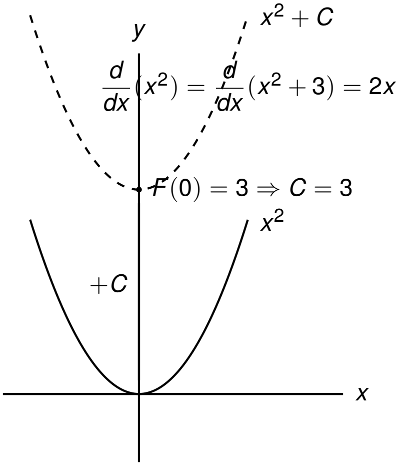
**Computation at scale**

- Iterative algorithms as matrix operations (e.g., gradient updates)
- Using sparsity/low-rank structure for fast estimation
- Quantitative text analysis and natural language processing

# Outline

# Why the constant of integration "$+C$"?

- **Indefinite integral = all antiderivatives.** $\int f(x)\,dx$ means the *family* $F(x) + C$ with $F'(x) = f(x)$.
- **Antiderivatives differ by a constant.** If $F'(x) = G'(x) = f(x)$ then $F(x) - G(x) = C$.
- **Definite integrals ignore $C$.** $[F(x) + C]_a^b = F(b) - F(a)$.
- **Initial condition picks $C$.** E.g. $F(0) = 3$ forces $C = 3$ when $\int 2x\,dx = x^2 + C$.



# Lorenz curve and the Gini index based on Stewart, J. (2012). Calculus (7th ed.)

**Lorenz curve** $L(x)$: income share of the bottom $x$ fraction of households.

- Domain $x \in [0,1]$, passes through $(0,0)$ and $(1,1)$.
- Increasing and convex; the line $y = x$ is *perfect equality*.
- The area between $y = x$ and $y = L(x)$ measures inequality.

**Gini index** $G$: twice the area between $y = x$ and $y = L(x)$,

$$G = 2\int_0^1 [x - L(x)]\,dx = 1 - 2\int_0^1 L(x)\,dx.$$

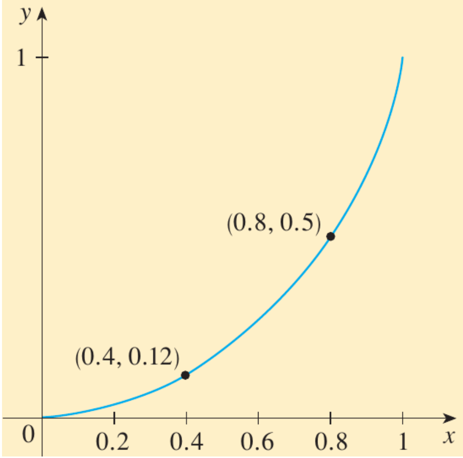



Figure: Example Lorenz curve with points $(0.4, 0.12)$ and $(0.8, 0.5)$.

# Lorenz curves: equality vs. inequality

- The 45° line $y = x$ represents perfect equality.
- Curves further from $y = x$ indicate more inequality (larger area $\Rightarrow$ larger $G$).
- All Lorenz curves pass through $(0, 0)$ and $(1, 1)$ and are concave up.

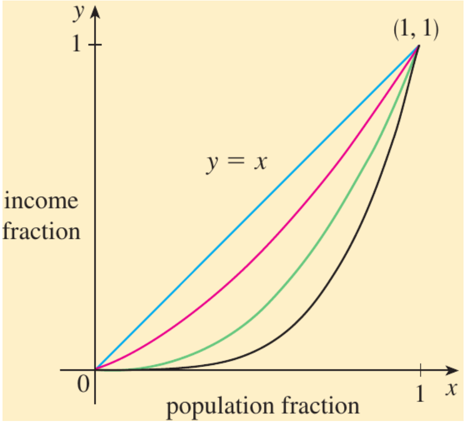



Figure: Families of Lorenz curves relative to $y = x$.

# Exercise 1 (a)–(b): Derive and interpret $G$

**(a) Show** $G = 2\int_0^1 (x - L(x))\,dx$.

- On $[0,1]$, the vertical gap between $y = x$ and $y = L(x)$ is $x - L(x)$.
- Area between the curves $= \int_0^1 (x - L(x))\,dx$.
- By definition, $G$ is twice that area:

$$\boxed{G = 2\int_0^1 (x - L(x))\,dx}.$$

Since $\int_0^1 x\,dx = \frac{1}{2}$, also $G = 1 - 2\int_0^1 L(x)\,dx$.

**(b) Extremes.**

- *Perfect equality:* $L(x) = x \Rightarrow G = 0$.
- *Perfect inequality:* $L(x) = 0$ for $x < 1$, jump to 1 at $x = 1 \Rightarrow \int_0^1 L(x)\,dx = 0 \Rightarrow \boxed{G = 1}$.

# Exercise 1: geometry

The shaded region shows the area between $y = x$ and $y = L(x)$ on $[0, 1]$.

Gini doubles this area, giving $G = 2\int_0^1 (x - L(x))\, dx$.

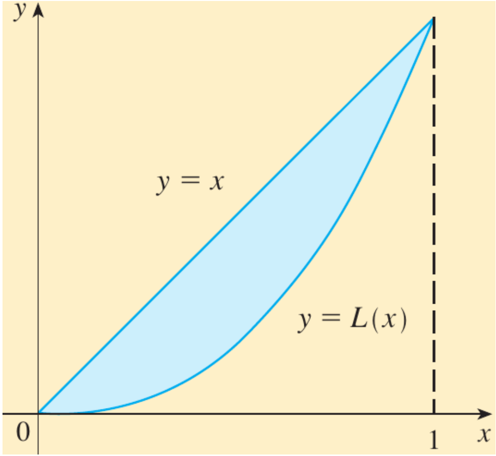



Figure: Area between $y = x$ and $y = L(x)$ (shaded).

# Exercise 2 (US 2008 data) - Computing Income Inequality

**Data (Lorenz points)**

| $x$ | 0 | 0.2 | 0.4 | 0.6 | 0.8 | 1 |
|---|---|---|---|---|---|---|
| $L(x)$ | 0 | 0.034 | 0.120 | 0.267 | 0.500 | 1 |

**(a) Richest** 20%**.** Top quintile $x \in [0.8, 1]$:

$$\text{share} = L(1) - L(0.8) = 1 - 0.5 = \boxed{50\%}.$$

**(b) Quadratic fit.** With $L(0) = 0$, least-squares $L(x) \approx ax^2 + bx$:

$$\boxed{a \approx 1.2311, \qquad b \approx -0.2731.}$$

**(c) Gini via the fit.**

$$\begin{aligned} \hat{G} &= 1 - 2\int_0^1 L(x)\,dx \\ &\approx 1 - 2\int_0^1 (ax^2 + bx)\,dx \\ &= 1 - 2\left(\frac{a}{3} + \frac{b}{2}\right) \\ &\approx 1 - 2\left(\frac{1.2311}{3} + \frac{-0.2731}{2}\right) \\ &= 1 - 2(0.2738) = \boxed{0.452}. \end{aligned}$$

(Uses $G = 2\int_0^1 (x - L(x))dx = 1 - 2\int_0^1 L(x)dx$.)

# Outline

# PS3 – Exercise 4 (iv): L'Hôpital

Compute $\lim_{y\to 0} \dfrac{e^y \sin y}{\log(1-y)}$.

**Step 1:** As $y \to 0$, $e^y \sin y \to 0$ and $\log(1-y) \to 0 \Rightarrow$ indeterminate $0/0$.
**Step 2:** Both functions are differentiable near 0 and the denominator's derivative is nonzero at 0.
**Step 3 (L'Hôpital):**

$$\lim_{y\to 0} \frac{e^y \sin y}{\log(1-y)} = \lim_{y\to 0} \frac{(e^y \sin y)'}{(\log(1-y))'} = \lim_{y\to 0} \frac{e^y \sin y + e^y \cos y}{-\frac{1}{1-y}}.$$

**Step 4:** Evaluate the (now continuous) limit:

$$-\lim_{y\to 0}(1-y)e^y\big(\sin y + \cos y\big) = -1 \cdot 1 \cdot (0+1) = \boxed{-1}.$$

(Series check: $e^y \sin y \sim y$, $\log(1-y) \sim -y$, ratio $\to -1$.)

# PS5 – Exercise 1 (iii): Definite integral

$$\int_2^4 e^y \, dy = e^y \Big|_2^4 = e^4 - e^2 = e^2(e^2 - 1) = \boxed{e^2(e-1)(e+1)}.$$

- FTC: an antiderivative of $e^y$ is $e^y$.
- Quick check: differentiate $e^y$ to recover the integrand.

# PS5 – Exercise 2 (iii): Indefinite Integral - Power rule with radicals

$$\int x\sqrt{x}\,dx = \int x \cdot x^{1/2}\,dx = \int x^{3/2}\,dx = \frac{x^{5/2}}{5/2} + C = \boxed{\frac{2}{5}x^{5/2} + C}.$$

- Rewrite $\sqrt{x}$ as $x^{1/2}$.
- Apply $\int x^n dx = \dfrac{x^{n+1}}{n+1} + C$ for $n \neq -1$ with $n = \frac{3}{2}$.

# Outline

# Continuity & Differentiability: Reminder

**Continuity at $a$.**

$$\lim_{x\to a} f(x) = f(a) \quad \text{i.e.} \quad \begin{cases} \text{(i) } f(a) \text{ defined,} \\ \text{(ii) } \lim_{x\to a} f(x) \text{ exists,} \\ \text{(iii) } \lim_{x\to a} f(x) = f(a). \end{cases}$$

**One–sided condition for *continuity* (esp. piecewise).**

$$\lim_{x\to a^-} f(x) \text{ and } \lim_{x\to a^+} f(x) \text{ exist and are equal to } f(a) \iff f \text{ is continuous at } a.$$

**Differentiability at $a$.**

$$f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h} \text{ exists (finite).}$$

**One–sided condition for *differentiability*.**

$$f'_-(a) = \lim_{h\to 0^-} \frac{f(a+h)-f(a)}{h}, \quad f'_+(a) = \lim_{h\to 0^+} \frac{f(a+h)-f(a)}{h},$$

$$f'(a) \text{ exists} \iff f'_-(a),\ f'_+(a) \text{ exist and } f'_-(a) = f'_+(a).$$

**To do:** Check $f(a)$ defined $\to$ left/right limits $\to$ continuity; if needed: left/right difference quotients $\to$ differentiability. $\boxed{\text{differentiability implies continuity}}$

# Exercise 1 - Continuity and differentiability

Consider the functions

$$f(x) = \begin{cases} x^2, & x \geq 0, \\ -x^2, & x < 0, \end{cases} \qquad g(x) = \begin{cases} x, & x > 1, \\ x^3, & x \leq 1. \end{cases}$$

With respect to continuity and differentiability, decide whether each statement is *True* or *False*:

(0) $f$ is continuous at $x = 0$.

(1) The derivative $f'$ is not continuous at $x = 0$.

(2) $g$ is differentiable at $x = 1$.

(3) The second derivative of $f$ is differentiable at $x = 0$.

(4) The function $h(x) = |f(x)|$ is not differentiable at $x = 0$.

# Solutions to (0) and (1)

**(0) "$f$ is continuous at $x = 0$"**

$$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} x^2 = 0, \qquad \lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} (-x^2) = 0, \qquad f(0) = 0.$$

Both one–sided limits equal $f(0)$, so $f$ is continuous at 0.

$$\boxed{\text{(0) True}}$$

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**(1) "$f'$ is not continuous at $x = 0$"**

$$f'(x) = \begin{cases} 2x, & x > 0, \\ -2x, & x < 0, \end{cases} \qquad f'(0) = \lim_{h\to 0} \frac{f(h) - f(0)}{h} = \lim_{h\to 0} \frac{\pm h^2}{h} = 0.$$

Thus

$$\lim_{x\to 0^+} f'(x) = \lim_{x\to 0^+} 2x = 0 = f'(0) = \lim_{x\to 0^-} (-2x) = \lim_{x\to 0^-} f'(x).$$

Hence $f'$ *is* continuous at 0.

$$\boxed{\text{(1) False}}$$

# Solutions to (2) and (3)

**(2) "$g$ is differentiable at $x = 1$"**

$$g(x) = \begin{cases} x, & x > 1, \\ x^3, & x \leq 1. \end{cases}$$

*One-sided derivatives at* 1*:*

$$g'_-(1) = \lim_{h \to 0^-} \frac{(1+h)^3 - 1}{h}$$

$$= \lim_{h \to 0^-} (3 + 3h + h^2) = 3.$$

$$g'_+(1) = \lim_{h \to 0^+} \frac{(1+h) - 1}{h} = 1.$$

Since $g'_-(1) \neq g'_+(1)$, $g'(1)$ does not exist.

$\boxed{\text{(2) False}}$

**(3) "$f''$ is differentiable at $x = 0$"**
*As seen in (1):*

$$f'(x) = \begin{cases} 2x, & x > 0, \\ -2x, & x < 0 \end{cases} = 2|x|, \qquad f'(0) = 0.$$

The function $f'(x) = 2|x|$ is *not* differentiable at $x = 0$. Therefore asking whether the *second* derivative of $f$ is differentiable at 0 does not even make sense: $f''(0)$ is not defined.

$\boxed{\text{(3) False}}$

# Solution to (4)

**(4) "$h(x) = |f(x)|$ is not differentiable at $x = 0$"**
For $x \geq 0$, $f(x) = x^2 \geq 0$; for $x < 0$, $f(x) = -x^2 \leq 0$. Hence

$$h(x) = |f(x)| = \begin{cases} |x^2| = x^2, & x \geq 0, \\ |-x^2| = x^2, & x < 0, \end{cases} \quad \Rightarrow \quad h(x) = x^2 \text{ for all } x \in \mathbb{R}.$$

The function $h(x) = x^2$ is differentiable for every $x \in \mathbb{R}$.

$$\boxed{\text{(4) False}}$$

# Exercise 2 (Sets)

Let $A$, $B$, $C$ be sets. The difference $A - B$ is $\{x : x \in A \text{ and } x \notin B\}$. Judge the statements:

(0) $(A \cup B) - C = (A - C) \cup (B - C)$, for any sets $A$, $B$, $C$.

(1) If $A - B = B - A$, then $A = B$.

# Solutions to Exercise 2

**(0)** $(A \cup B) - C = (A - C) \cup (B - C)$.
*True*. Use $X - Y = X \cap Y^c$ and distributivity:

$$(A \cup B) - C = (A \cup B) \cap C^c = (A \cap C^c) \ \cup \ (B \cap C^c) = (A - C) \ \cup \ (B - C).$$

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**(1)** If $A - B = B - A$, then $A = B$.
*False*. Counterexample with two nonempty disjoint sets. Take $A = (-1, 0)$ and $B = (1, 2)$. Then

$$A - B = A, \qquad B - A = B, \qquad \text{so } A - B \neq B - A,$$

yet (being disjoint) we can also pick *any* nonempty disjoint $A$, $B$ and have $A - B = A$ and $B - A = B$, which are equal only if $A = B = \varnothing$. Thus equality of differences need not imply $A = B$.

# Exercise 3: Unconstrained optimization (one variable)

Find and classify the critical points of

$$f(x) = x^3 - 3x^2 + 2.$$

Indicate which are local minima/maxima and give the function values at those points.

# Solution to Exercise 3

**1) Critical points (FOC)** Solve $f'(x) = 0$.

$$f'(x) = 3x^2 - 6x = 3x(x-2) \quad \Rightarrow \quad x = 0,\ x = 2.$$

**2) Second–derivative or sign test (SOC)**

$$f''(x) = 6x - 6.$$

$f''(0) = -6 < 0 \ \Rightarrow\ $ local maximum at $x = 0$, $\qquad f''(2) = 6 > 0 \ \Rightarrow\ $ local minimum at $x = 2$

**3) Values at the extrema.**

$$f(0) = 2 \quad (\text{local max}), \qquad f(2) = 8 - 12 + 2 = -2 \quad (\text{local min}).$$

**4) Global behavior.** As $x \to +\infty$, $f(x) \to +\infty$; as $x \to -\infty$, $f(x) \to -\infty$. Hence no global maximum or minimum on $\mathbb{R}$; only the local extrema above.